

Assignment 3

Database Integration in a Web Application

Objective:

Connect a web application to a database, create the required tables or collections, and implement backend logic for data movement in and out of the database.

Assignment Requirements

1. **Web Application Setup:**

Set up the web application.

Follow a clean **MVC (Model-View-Controller)** project structure.

2. **Database Integration:**

Connect your web application to database (depending on your choice).

Install the necessary libraries.

All connection parameters (database URI, secret keys, ports, etc.) must be stored in a `.env` file.

3. **Create Tables or Collections:**

Design and create all necessary **tables (SQL)** or **collections (NoSQL)** for your application.

Examples:

User Collection/Table: Fields `id`, `name`, `email`, `password`, `role` (admin or user).

Product Collection/Table (if applicable): Fields `productId`, `name`, `description`, `price`, `quantity`.

Order Collection/Table (if applicable): Fields `orderId`, `userId`, `totalAmount`, `orderStatus`, `createdAt`, `updatedAt`

4. **Routes and Backend Logic:**

Implement the following routes and backend logic.

Examples:

User Registration:

Method: `POST /api/register`

Saves user data in the database after hashing the password.

User Authentication:

Method: `POST /api/login`

Verifies user credentials and returns a JWT (JSON Web Token).

Add Product (if applicable):

Method: `POST /api/products`

Adds a new product to the database.

5. **Environment Variables File (`.env`):**

Store all sensitive information in the `.env` file.

Example:

```
PORT=3000
MONGODB_URI=mongodb://localhost:27017/yourdbname
JWT_SECRET=your_secret_key
```

6. **Validation and Error Handling:**

Add middleware to validate input data (e.g., password length, email format, etc.).

Implement error handling for invalid requests, database errors, and unauthorized access.

7. **Security Measures:**

Use `bcrypt` to hash passwords before storing them.

Implement JWT authentication to secure protected routes.

Ensure that sensitive data is not exposed in API responses.